

Jonathan Nguyen

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Education

Chalmers University of Technology — MSc, Engineering Mathematics & Computational Science Aug 2026 – Jun 2028 (Expected)

- Master's programme within the Engineering Physics degree; specialization in computational mathematics, optimization, and statistical learning.

Chalmers University of Technology — BSc, Engineering Physics Aug 2023 – Jun 2026

- GPA: 4.83/5.0
- Additional computer science and higher-level mathematics coursework at the University of Gothenburg, taken alongside the degree, equivalent to 150-200% of studies.
- **Relevant coursework:** Probability & Statistics, Numerical analysis (Python), Linear Algebra, Machine learning & Bayesian Inference, Data structures & Algorithms, Introduction to computer engineering.

Experience

Software Engineer Intern, Balance — Stockholm, Sweden Jun 2026 – Aug 2026

- Designed and deployed an AI-driven contract engine using Claude on AWS Bedrock to parse Swedish legal documents into schema-validated templates via a two-pass, automated verification pipeline.
- Productionized the generation workflow using asynchronous Celery workers, Jinja2 + WeasyPrint for live HTML-to-PDF rendering, and strict workspace-scoped access control.
- Engineered enterprise full-stack features (FastAPI, React/TS, PostgreSQL), including an automated accounting checklist and an admin analytics dashboard with time-bucketed SQL and eliminated N+1 queries.

Data Scientist Intern, Algeno — Gothenburg, Sweden Feb 2026 – Present

- Built a time-series anomaly-detection engine for indoor heating sensors over 4.6M readings, implementing robust statistical detectors (median/MAD z-scores, least-squares drift, derivative and cross-sensor checks) and a nonparametric dynamic threshold (Hundman et al., 2018), fully vectorized in pandas for performance at scale.
- Extracted, cleaned, and modeled sensor time series for temperature forecasting and sensor segmentation, applying scikit-learn clustering (KMeans, DBSCAN, PCA, silhouette scoring) and validating model selection on held-out data.
- Shipped the detection engine as a tested Python package (pytest) with a React visualization frontend for exploring flagged anomalies across sensors.

Private Teacher, Leexer — Gothenburg, Sweden Feb 2023 – Feb 2026

- Tutored students aged 14–19 in physics and mathematics, planning lessons and adapting explanations to each student's level.

Projects

GPU Inference Engine — GPT-2 from scratch in CUDA In progress

- Building a GPT-2 (124M) inference engine from scratch in CUDA, hand-writing every kernel (layernorm, softmax, attention, GELU, matmul) and verifying each layer's activations against a HuggingFace reference to $<1e-4$.
- Wrote and profiled memory-bound CUDA kernels on a T4, reasoning from the roofline model; achieved 260 GB/s on a bandwidth-bound kernel (81% of the card's 320 GB/s peak).
- Quantified the cost of uncoalesced memory access by controlled experiment — a $2.9\times$ bandwidth drop from access pattern alone — and used the result to guide kernel memory layout.

Skills

Programming Languages: Python, TypeScript/JavaScript, SQL (PostgreSQL), C, Java, HTML/CSS

Frameworks & Libraries: pandas, NumPy, scikit-learn, scipy, Matplotlib, FastAPI, SQLAlchemy, React, pytest

Tools & Platforms: Git/GitHub, AWS (Bedrock, S3), Celery, Alembic, Vite, TanStack Query, CAD

Languages: Swedish (native), English (fluent), Vietnamese